

BOM Site Handover Checklist

Quote: EP-2025-00028

Customer: Madam Pearl

Project: Madam Pearl — Residential Hybrid (2 options)

Option: OPTION 1

System: hybrid — 5.09 kW, 9 panels

Date: 15 May 2026

Core equipment

☐	Description	Qty	Notes
	570W Longi Solar Panel	8.0	
	6.5 kW Hybrid Inverter	1.0	
	5 kWh Lithium Ion Battery	2.0	
	Mounting structure (rails) (proforma rail sets)	5.0	
	PV mid / end clamp (each) (3.5× panels)	28.0	
	MC4 connector pair (~1.2 pairs per panel)	10.0	
	RR 10MM Copper Cable (100m coil) (AC feeder coils vs inverter kW)	2.0	
	Battery cable 35mm copper (per meter) (~2 m per kWh)	20.0	
	AC isolator switch 40A (AC isolator per inverter)	1.0	
	100A changeover (1 per inverter / genset path)	1.0	
	250A MCCB battery breaker (battery bank protection)	2.0	

Balance of system (catalog BOM)

☐	Item	Qty	SKU / usage
	AC isolator switch 40A	1.0	BOS-AC-ISO-40A — AC side disconnect at inverter for safe maintenance.
	DC isolator switch 32A	1.0	BOS-DC-ISO-32A — Lockable DC disconnect for maintenance — typically one per string or per inverter DC input.
	Earthing kit (rod, clamp, 16mm earth cable)	2.0	BOS-EARTH-KIT — Equipment bonding and array grounding per local wiring rules.
	DC/AC labeling & warning kit	1.0	BOS-LABEL-KIT — Circuit labels, arc-flash / DC warning stickers for commissioning sign-off.
	MC4 connector pair	10.0	BOS-MC4-PAIR — DC plug connections at modules and string ends; ~1 pair per module interconnection.
	PV mid / end clamp (each)	28.0	BOS-PV-CLAMP-EA — Secures modules to rails (mid + end clamps).
	PVC trunking 25mm (per metre)	12.0	BOS-TRUNK-25MM-M — Cable management for AC/DC runs on wall/facade.

	13-way breaker box	2.0	BOX-DB-13WAY — Main AC DB for breakers, SPD, and feeds.
	6-way breaker box	1.0	BOX-DB-6WAY — Small AC distribution enclosure.
	AC cable 6 mm ² (100m coil)	1.0	CAB-AC-6MM-100M — Branch AC wiring, DB to loads (smaller runs); supplements RR10 where needed.
	Battery cable 35mm copper (per meter)	20.0	CAB-BATT-35MM-M — Battery interconnect and inverter-battery DC runs (hybrid/off-grid).
	Single core 4mm PV cable (100m coil)	2.0	CAB-PV-4MM-100M — String homerun / interconnection for residential arrays (≤30 panels, ≤20 kW).
	RR 10MM Copper Cable (100m coil)	2.0	CAB-RR10MM-100M — AC feeder from inverter to DB / changeover (heavy residential & commercial).
	Miscellaneous (MC4, trunking, PVC, glands, fixings, lugs, ties)	1.0	KIT-MISC-BOS — Glands, lugs, cable ties, small fixings not itemized separately.
	Mounting structure (rails)	5.0	MNT-RAIL-SET-350 — Roof rail segments + typical fixings per set — matches Ghana proforma 'Mounting Structure (Rails)'.
	100A AC breaker	1.0	PRT-AC100-1P — AC overcurrent protection at DB / inverter output.
	100A changeover	1.0	PRT-CO-100A — Grid / generator / inverter selection (hybrid & backup).
	63A DC MCB 2P	2.0	PRT-DC63-2P — DC string / combiner protection (one per string group).
	250A MCCB battery breaker	2.0	PRT-MCCB-250A — Battery bank main overcurrent protection.
	SPD / lightning arrester (500V class)	1.0	PRT-SPD-500V — Lightning / surge protection on DC side (standard residential).
	Installation & commissioning (standard)	1.0	SRV-INSTALL-6000 — Labour, testing, handover — tier ≤25 kW.
	Transportation of logistics	1.0	SRV-TRANSPORT-STD — Delivery of equipment to site (Ghana proforma reference).

Site / manual items (confirm on site)

☐	Item	Notes
	Engineering / design drawings	String layout, single-line diagram, load study — not auto-generated.
	ECG / utility application fees	Grid-tied export or net-metering approvals vary by region.
	Scaffolding / height access	Multi-storey or steep roof — site-specific.
	Concrete ballast / ground mount civils	Ground-mount only; not in roof-mount BOM.
	Generator integration panel	Add when changeover feeds existing gen set beyond standard CO.
	Monitoring / Wi-Fi dongle	Inverter brand-specific; add from equipment catalog.

Commissioning sign-off

Installer / lead technician

Client representative

Date completed

Energy Precisions QC

Generated from sizing-driven BOM. Tick each line when delivered to site and installed. Final string design (VD, Isc) remains the supervising engineer's responsibility.